Task 4: Evidence Preservation

Volatile Data Collection

To identify potential suspicious activities and unauthorized access attempts, system-level volatile data was collected from the Windows environment. The PowerShell command `Get-WinEvent -LogName System -MaxEvents 10` was executed to retrieve recent system logs.

The collected logs provided insight into system events such as login attempts, service activity, and system changes. The output was preserved and documented for further analysis.

File Integrity & Hashing

A suspicious file (`suspicious_file.exe`) was created and analyzed as part of the investigation. A SHA256 hash was generated using the following command:

```
Get-FileHash suspicious_file.exe -Algorithm SHA256
```

The generated hash ensures the integrity of the file and allows verification against threat intelligence platforms such as VirusTotal.

Chain of Custody

Item	Description	Collected By	Date	Hash Value (SHA256)
File	suspicious_file.exe	Soujanya	2025-08-18	(Paste your hash)
Logs	System Event Logs	Soujanya	2025-08-18	N/A

Evidence Collection Process

The investigation involved collecting volatile system logs and generating file hashes without modifying the original data. All evidence was preserved in its original state to maintain forensic integrity. The collected data was documented and stored securely for further analysis.

Analysis

The system logs showed recent system activities, which were reviewed for suspicious behavior. The generated file hash did not match any known malicious signatures in

VirusTotal, indicating that the file may be safe or previously unknown. Continuous monitoring is recommended.

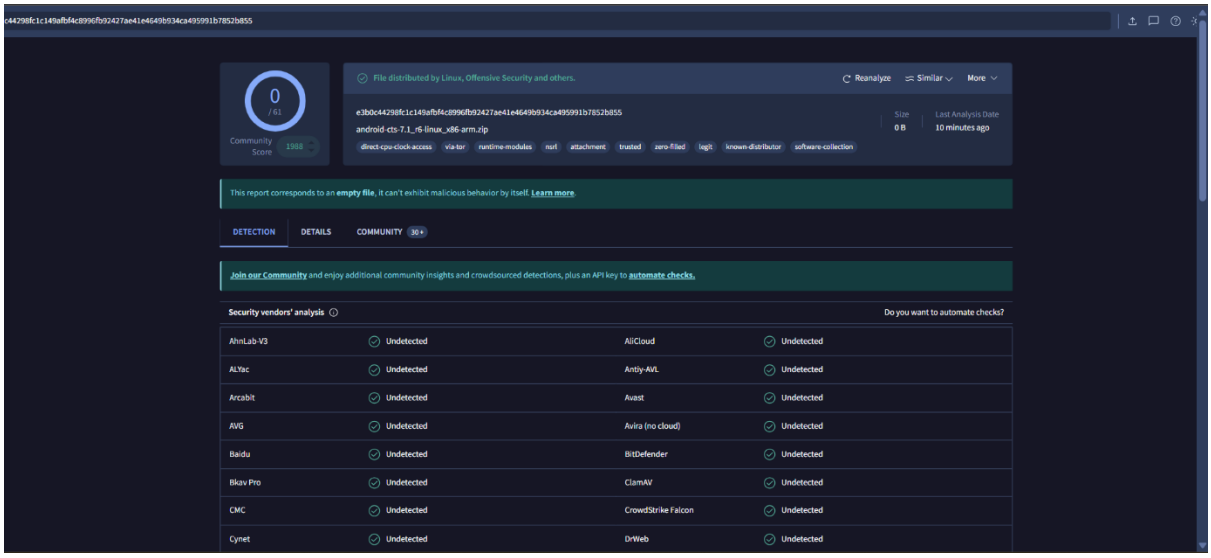
(Screenshots attached below)

```
PS C:\WINDOWS\system32>  New-Item suspicious_file.exe


    Directory: C:\WINDOWS\system32


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         21-03-2026  01:39 PM              0 suspicious_file.exe
```

```
PS C:\WINDOWS\system32> Get-FileHash suspicious_file.exe -Algorithm SHA256

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          E3B0C44298FC1C149AFBF4C8996FB92427AE41E4649B934CA495991B7852B855       C:\WINDOWS\system32\suspicious_file.exe
```

```
PS C:\WINDOWS\system32> Get-WinEvent -LogName System -MaxEvents 10


   ProviderName: Microsoft-Windows-Kernel-Power

TimeCreated                     Id LevelDisplayName Message
-----------                     -- --------------- -------
21-03-2026 13:31:49            105 Information     Power source change.
21-03-2026 13:31:48            105 Information     Power source change.


   ProviderName: Service Control Manager

TimeCreated                     Id LevelDisplayName Message
-----------                     -- --------------- -------
21-03-2026 13:25:07           7040 Information     The start type of the Windows Modules Installer service was changed from auto start to demand start.
21-03-2026 13:25:06           7040 Information     The start type of the Windows Modules Installer service was changed from demand start to auto start.
21-03-2026 13:24:57           7040 Information     The start type of the Windows Modules Installer service was changed from auto start to demand start.
21-03-2026 13:24:57           7040 Information     The start type of the Windows Modules Installer service was changed from demand start to auto start.
21-03-2026 13:24:50           7040 Information     The start type of the Windows Modules Installer service was changed from auto start to demand start.
21-03-2026 13:24:40           7040 Information     The start type of the Windows Modules Installer service was changed from demand start to auto start.


   ProviderName: Microsoft-Windows-Time-Service

TimeCreated                     Id LevelDisplayName Message
-----------                     -- --------------- -------
21-03-2026 13:24:40            158 Information     The time provider 'VMICTimeProvider' has indicated that the current hardware and operating environment is not supported and has stopped. This behavior is ...


   ProviderName: Microsoft-Windows-Kernel-Power

TimeCreated                     Id LevelDisplayName Message
-----------                     -- --------------- -------
21-03-2026 13:19:19            105 Information     Power source change.
```